CSA0404-OPERATING SYSTEMS

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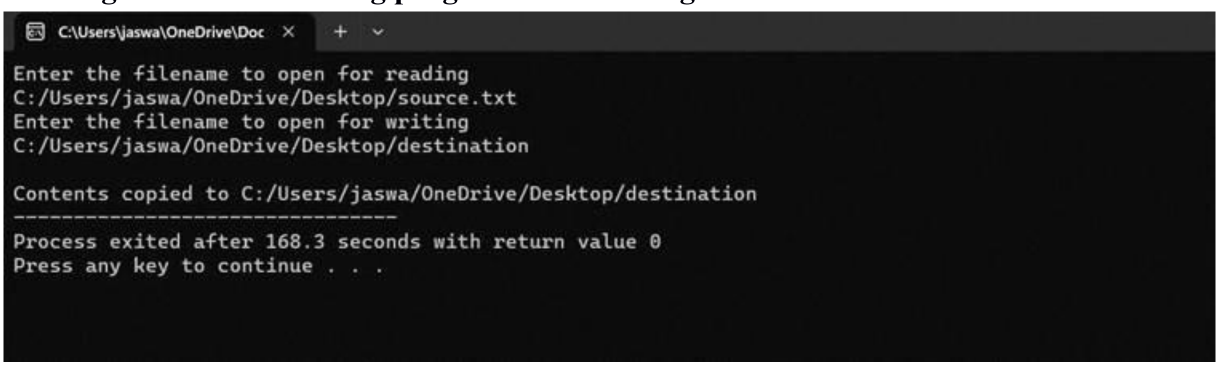
3. Design a CPU scheduling program with C using First Come First Served



technique with the following considerations.

a. **All processes are activated at time 0.**
b. **Assume that no process waits on I/O devices.**

AIM: Design a CPU scheduling program with C using First Come First Served technique with the following considerations.

a. All processes are activated at t ime 0.
b. Assume that no process waits on I/O devices.

Given:-

```c
int main() {
    int n, i;
    int bt[20], wt[20], tat[20];
    float avg_wt = 0, avg_tat = 0;

    printf("Enter the number of processes: ");
    scanf("%d", &n);

    printf("Enter Burst Time:\n");
    for(i = 0; i < n; i++) {
        printf("P%d: ", i + 1);
        scanf("%d", &bt[i]);
    }
    wt[0] = 0;

    for(i = 1; i < n; i++) {
        wt[i] = wt[i - 1] + bt[i - 1];
    }
    for(i = 0; i < n; i++) {
        tat[i] = wt[i] + bt[i];
        avg_wt += wt[i];
        avg_tat += tat[i];
    }
    printf("\nProcess\tBurst Time\tWaiting Time\tTurnaround Time\n");
    for(i = 0; i < n; i++) {
        printf("P%d\t%d\t\t%d\t\t%d\n",
         i + 1, bt[i], wt[i], tat[i]);
    }

    printf("\nAverage Waiting Time = %.2f", avg_wt / n);
    printf("\nAverage Turnaround Time = %.2f\n", avg_tat / n);
    return 0;
}
```

Output:

```
Enter the number of processes: 3
Enter Burst Time:
P1: 12
P2: 14
P3: 16

Process Burst Time      Waiting Time    Turnaround Time
P1      12              0               12
P2      14              12              26
P3      16              26              42

Average Waiting Time = 12.67
Average Turnaround Time = 26.67


...Program finished with exit code 0
Press ENTER to exit console.
```

4. Construct a scheduling program with C that selects the waiting process with the smallest execution time to execute next.

Given:-

```c
#include<stdio.h>
int main()
{
    int bt[10], wt[10], tat[10], p[10];
    int i, j, n, temp;

    printf("Enter number of processes: ");
    scanf("%d", &n);

    for(i=0;i<n;i++) {
        printf("P%d Burst Time: ", i+1);
        scanf("%d",&bt[i]);
        p[i]=i+1;
    }

    for(i=0;i<n-1;i++)
        for(j=i+1;j<n;j++)
            if(bt[i]>bt[j]) {
                temp=bt[i]; bt[i]=bt[j]; bt[j]=temp;
                temp=p[i]; p[i]=p[j]; p[j]=temp;
            }

    wt[0]=0;
    for(i=1;i<n;i++)
        wt[i]=wt[i-1]+bt[i-1];

    printf("\nP\tBT\tWT\tTAT\n");
    for(i=0;i<n;i++) {
        tat[i]=bt[i]+wt[i];
        printf("P%d\t%d\t%d\t%d\n",p[i],bt[i],wt[i],tat[i]);
    }

    return 0;
}
```

Output:-

```
Enter number of processes: 4
P1 Burst Time: 3
P2 Burst Time: 3
P3 Burst Time: 4
P4 Burst Time: 4

P       BT      WT      TAT
P1      3       0       3
P2      3       3       6
P3      4       6       10
P4      4       10      14

...Program finished with exit code 0
Press ENTER to exit console.
```

5.Construct a scheduling program with C that selects the waiting process with the highest priority to execute next.

Given:-

```c
#include <stdio.h>

int main() {
    int n, i, j, bt[10], pr[10], p[10], wt[10], tat[10], temp;

    printf("Enter number of processes: ");
    scanf("%d", &n);

    for(i = 0; i < n; i++) {
        printf("P%d Burst Time: ", i + 1);
        scanf("%d", &bt[i]);
        printf("P%d Priority: ", i + 1);
        scanf("%d", &pr[i]);
        p[i] = i + 1;
    }

    for(i = 0; i < n - 1; i++)
        for(j = i + 1; j < n; j++)
            if(pr[i] > pr[j]) {
                temp = pr[i]; pr[i] = pr[j]; pr[j] = temp;
                temp = bt[i]; bt[i] = bt[j]; bt[j] = temp;
                temp = p[i]; p[i] = p[j]; p[j] = temp;
            }

    wt[0] = 0;
    for(i = 1; i < n; i++)
        wt[i] = wt[i - 1] + bt[i - 1];

    printf("\nP\tBT\tPR\tWT\tTAT\n");
    for(i = 0; i < n; i++) {
        tat[i] = wt[i] + bt[i];
        printf("P%d\t%d\t%d\t%d\t%d\n", p[i], bt[i], pr[i], wt[i], tat[i]);
    }

    return 0;
}
```

Output:-

```
Enter number of processes: 3
P1 Burst Time: 9
P1 Priority: 2
P2 Burst Time: 9
P2 Priority: 3
P3 Burst Time: 4
P3 Priority: 2

P       BT      PR      WT      TAT
P1      9       2       0       9
P3      4       2       9       13
P2      9       3       13      22


...Program finished with exit code 0
Press ENTER to exit console.
```